

Bitwise differential trail through **WARP** found by SAT

CiVerLy

August 20, 2026

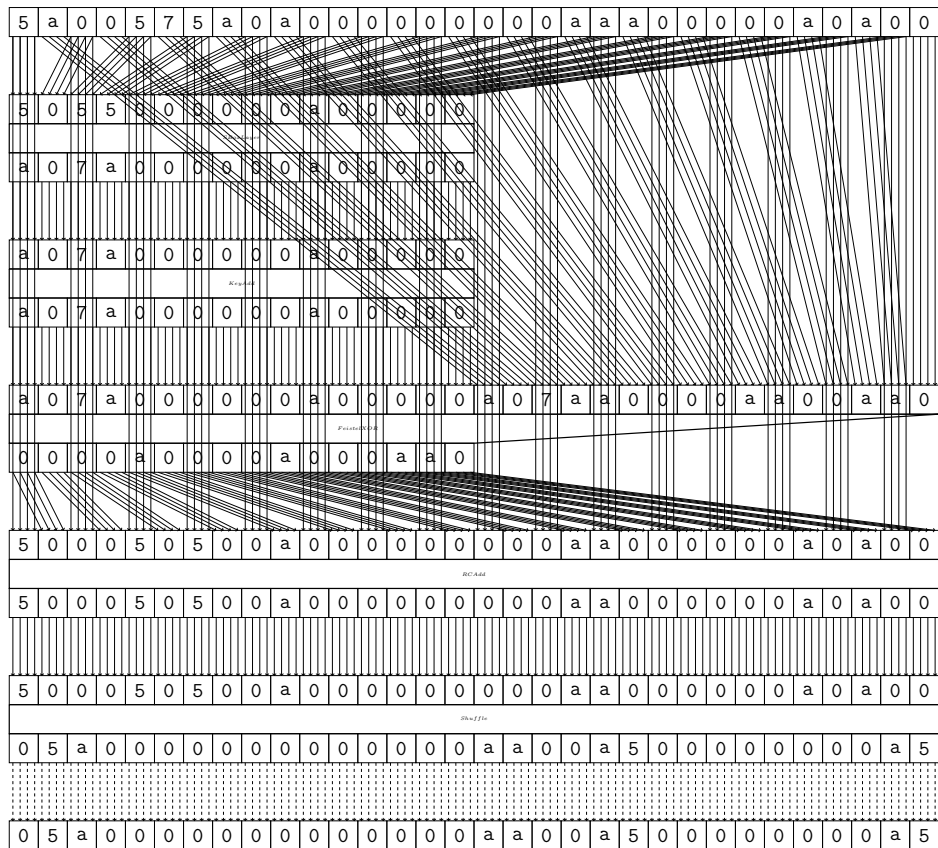
1 WARP

Maximal differential probability: 2^{-56}

5	a	0	0	5	7	5	a	0	a	0	0	0	0	0	0	0	0	a	a	a	0	0	0	0	a	0	a	0	0				
integrated																																	
0	5	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	a	0	0	a	5	0	0	0	0	0	0	0	a	5		
0	5	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	a	0	0	a	5	0	0	0	0	0	0	0	0	a	5	
integrated																																	
0	0	0	0	0	0	5	a	0	0	0	0	0	0	0	0	0	a	a	0	a	0	0	0	0	0	0	0	0	0	0	a	0	
0	0	0	0	0	0	5	a	0	0	0	0	0	0	0	0	0	a	a	0	a	0	0	0	0	0	0	0	0	0	0	0	a	0
integrated																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	5	a	a	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	5	a	a	0	0	0	0	0	0	0	0	0	
integrated																																	
0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
integrated																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	
integrated																																	
0	0	0	0	a	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	a	0	0	0	0	0	
0	0	0	0	a	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	a	0	0	0	0	0	
integrated																																	
0	a	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	a	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
integrated																																	
5	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
5	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
integrated																																	
0	0	0	0	0	0	7	0	a	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	5	
0	0	0	0	0	0	7	0	a	0	0	0	0	0	0	0	0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	5	
integrated																																	
0	0	a	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	5	a	a	0	0	0	0	0	0	
0	0	a	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	5	a	a	0	0	0	0	0	0	
integrated																																	
0	0	a	0	0	0	0	0	0	a	0	0	0	0	a	0	0	0	5	5	0	0	0	0	0	0	5	7	a	0	0	0	0	
0	0	a	0	0	0	0	0	0	a	0	0	0	0	a	0	0	0	5	5	0	0	0	0	0	0	5	7	a	0	0	0	0	
integrated																																	
0	0	a	0	0	0	0	0	0	7	5	0	0	0	5	0	5	5	0	0	f	0	0	a	0	0	0	0	7	a	5	0	0	
integrated																																	
0	0	a	0	0	0	0	0	0	7	5	0	0	0	5	0	5	5	0	0	f	0	0	a	0	0	0	0	7	a	5	0	0	

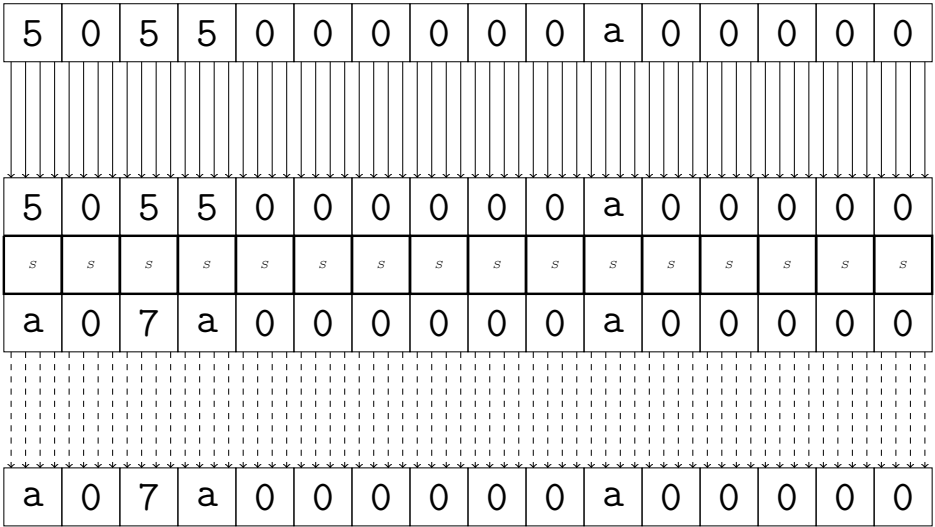
2 warp_round

Maximal differential probability: $2^{-8.0}$



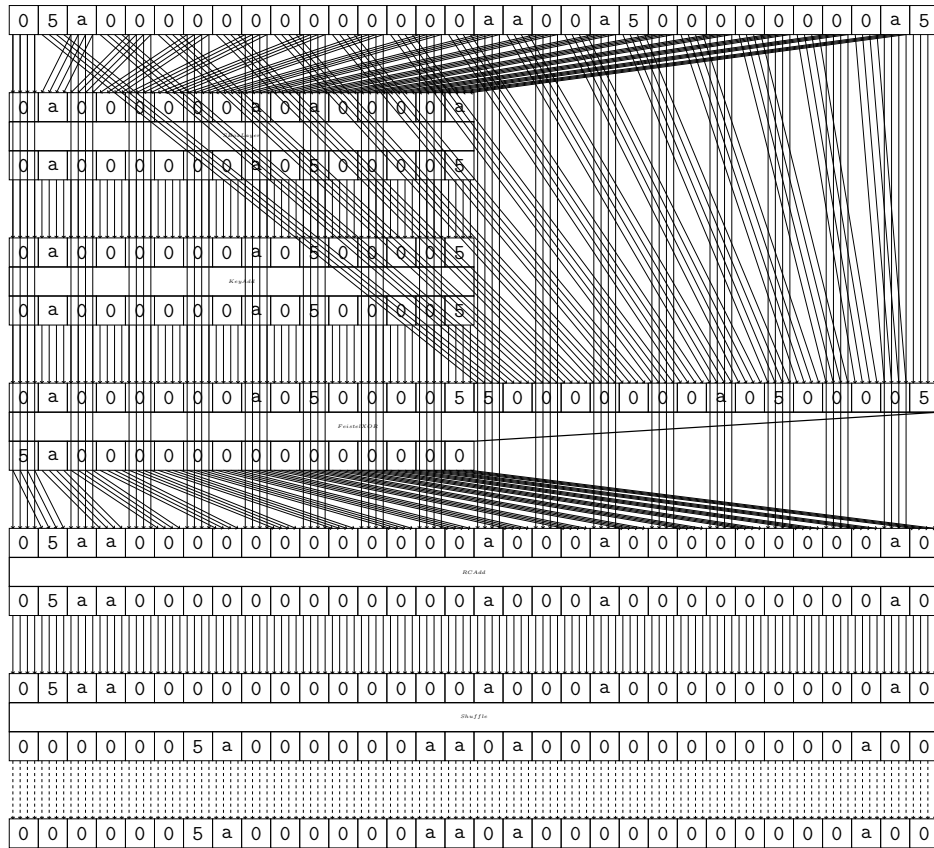
3 SBoxLayer

Maximal differential probability: $2^{-8.0}$



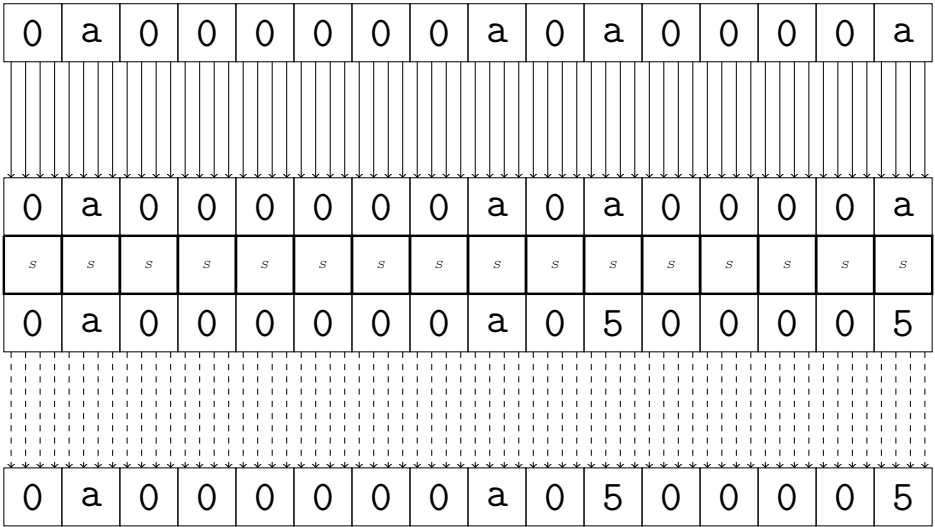
4 warp_round

Maximal differential probability: $2^{-8.0}$



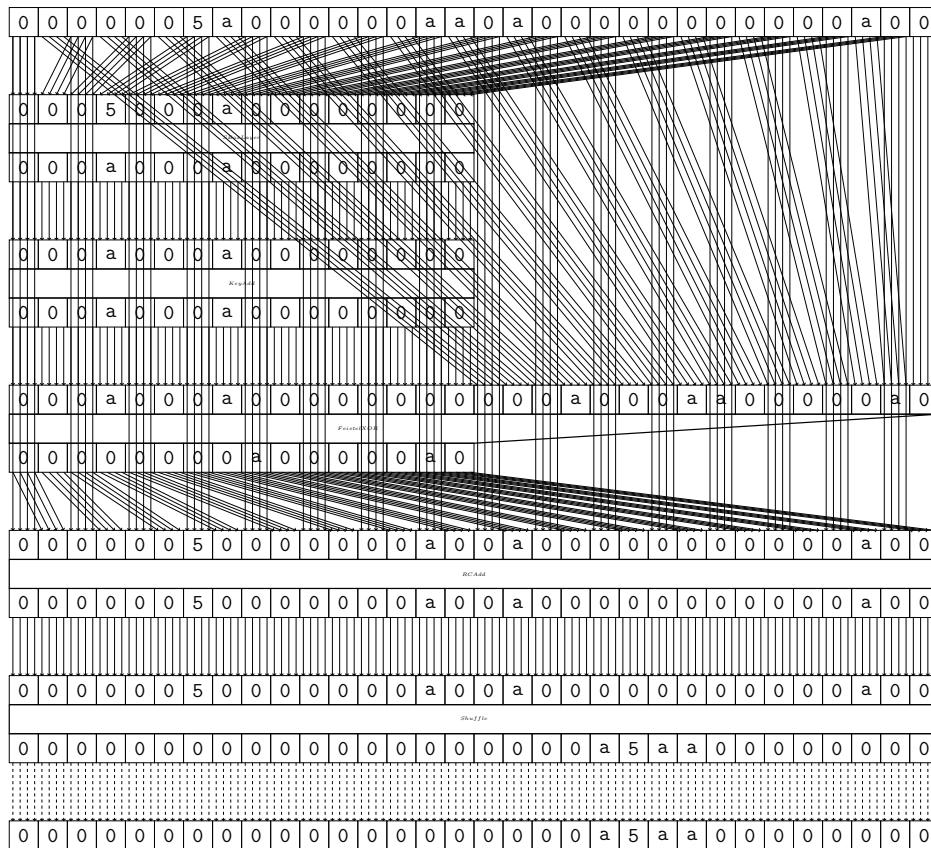
5 SBoxLayer

Maximal differential probability: $2^{-8.0}$



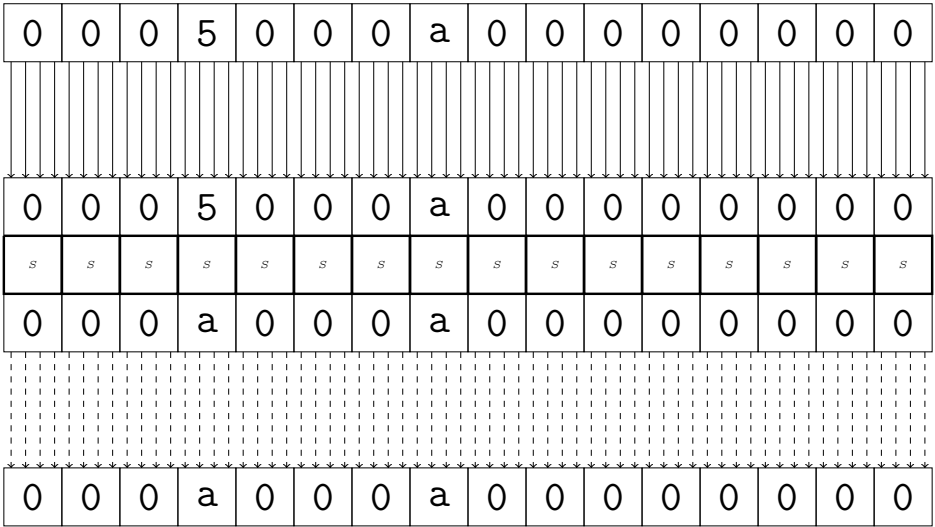
6 warp_round

Maximal differential probability: $2^{-4.0}$



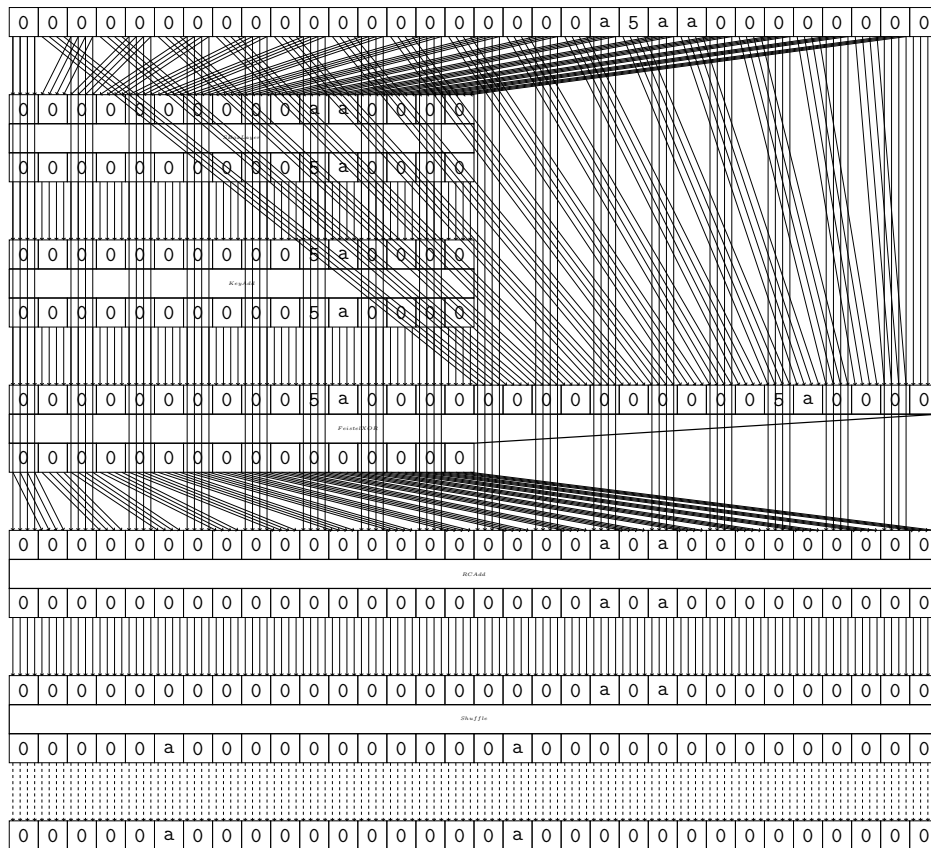
7 SBoxLayer

Maximal differential probability: $2^{-4.0}$



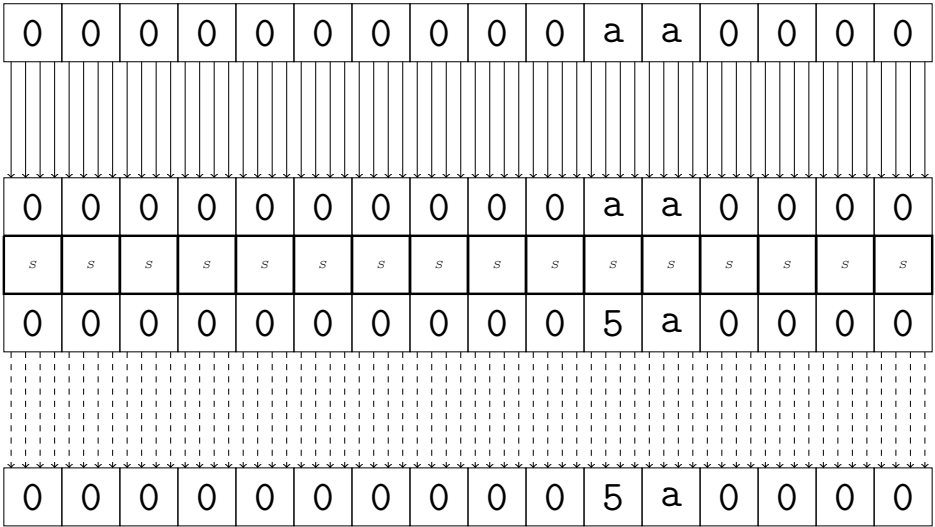
8 warp_round

Maximal differential probability: $2^{-4.0}$



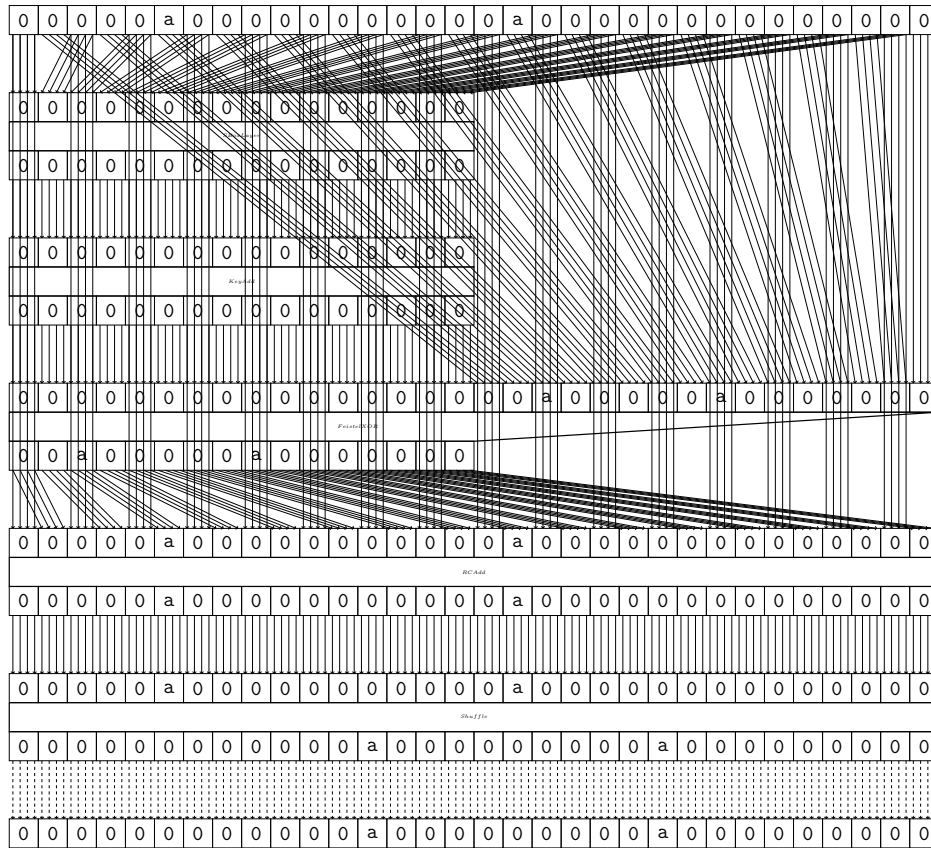
9 SBoxLayer

Maximal differential probability: $2^{-4.0}$



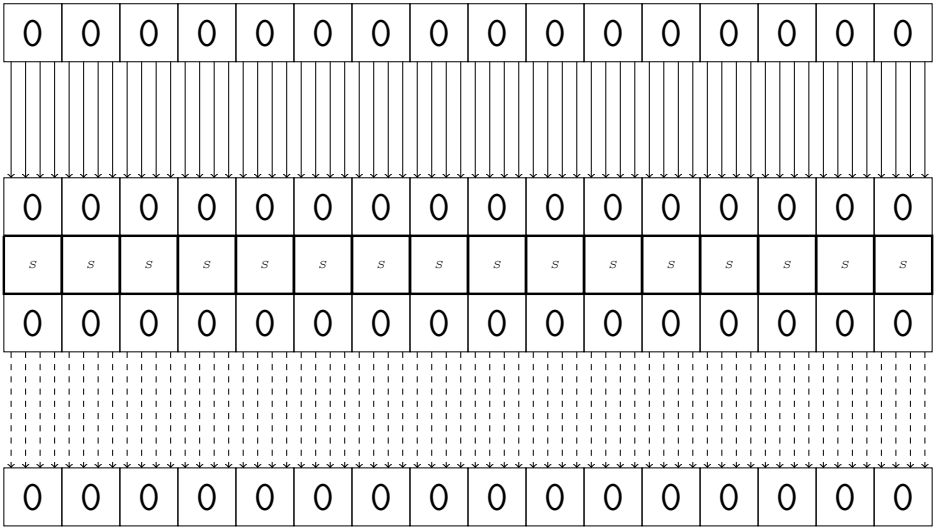
10 warp_round

Maximal differential probability: $2^{-0.0}$



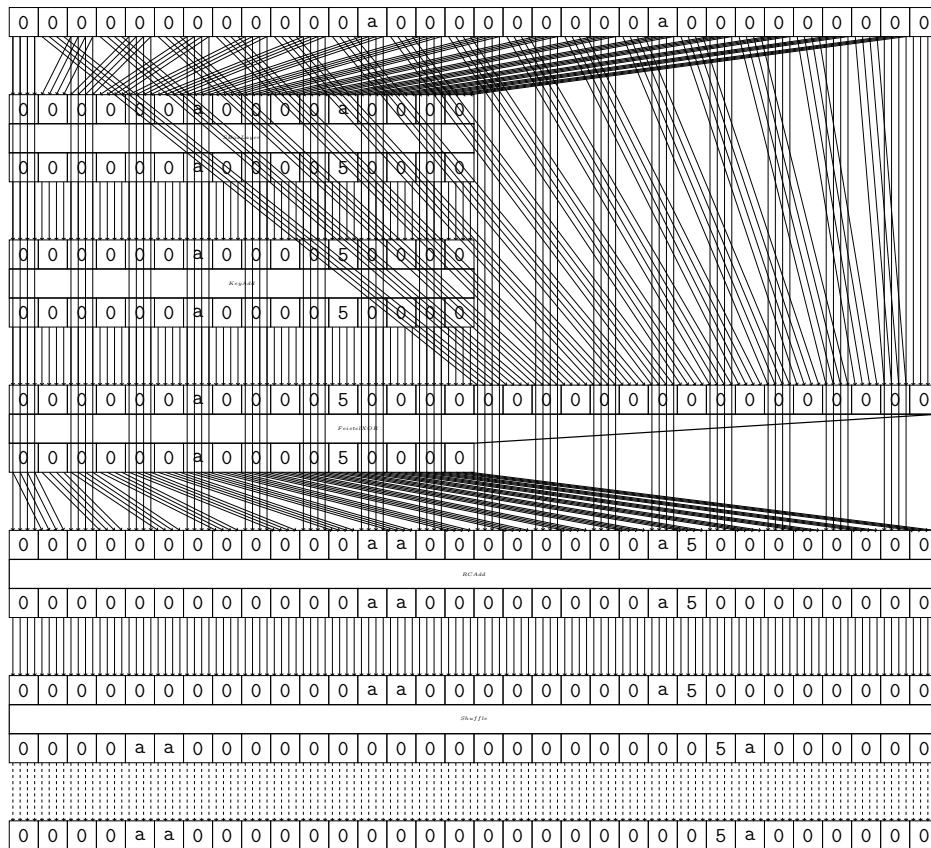
11 SBoxLayer

Maximal differential probability: $2^{-0.0}$



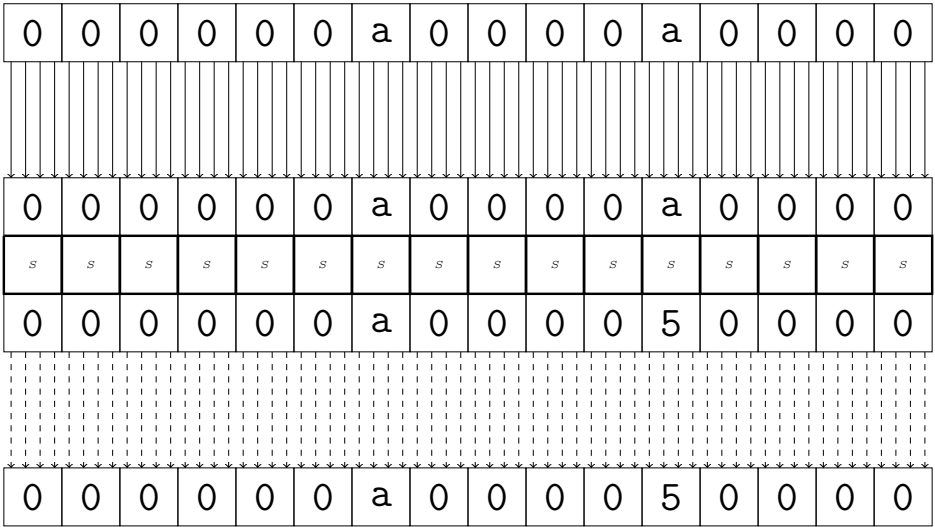
12 warp_round

Maximal differential probability: $2^{-4.0}$



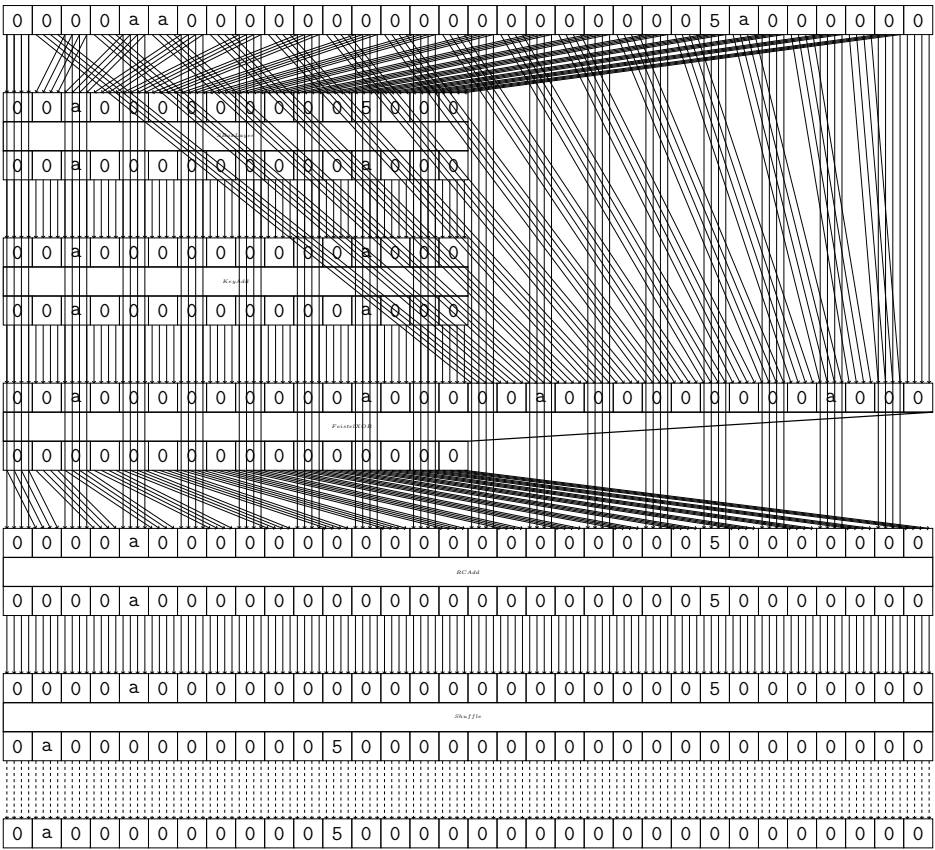
13 SBoxLayer

Maximal differential probability: $2^{-4.0}$



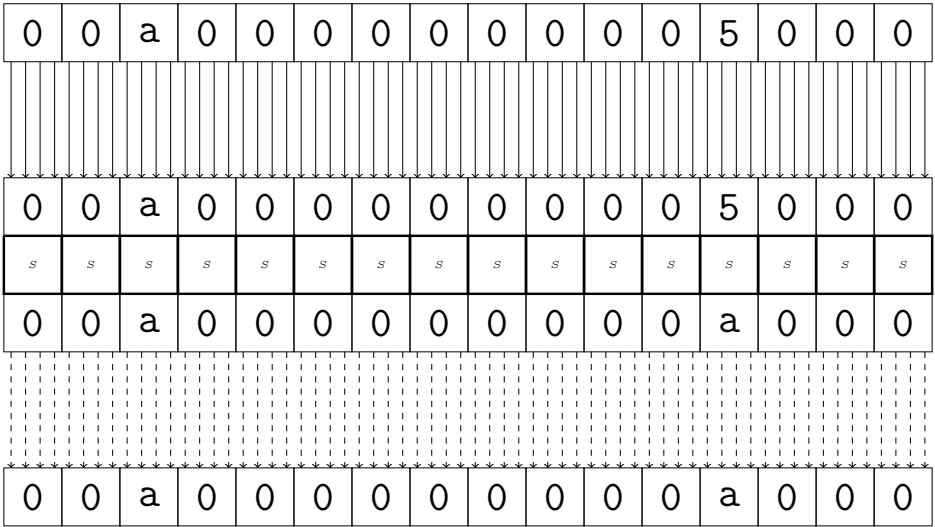
14 warp_round

Maximal differential probability: $2^{-4.0}$

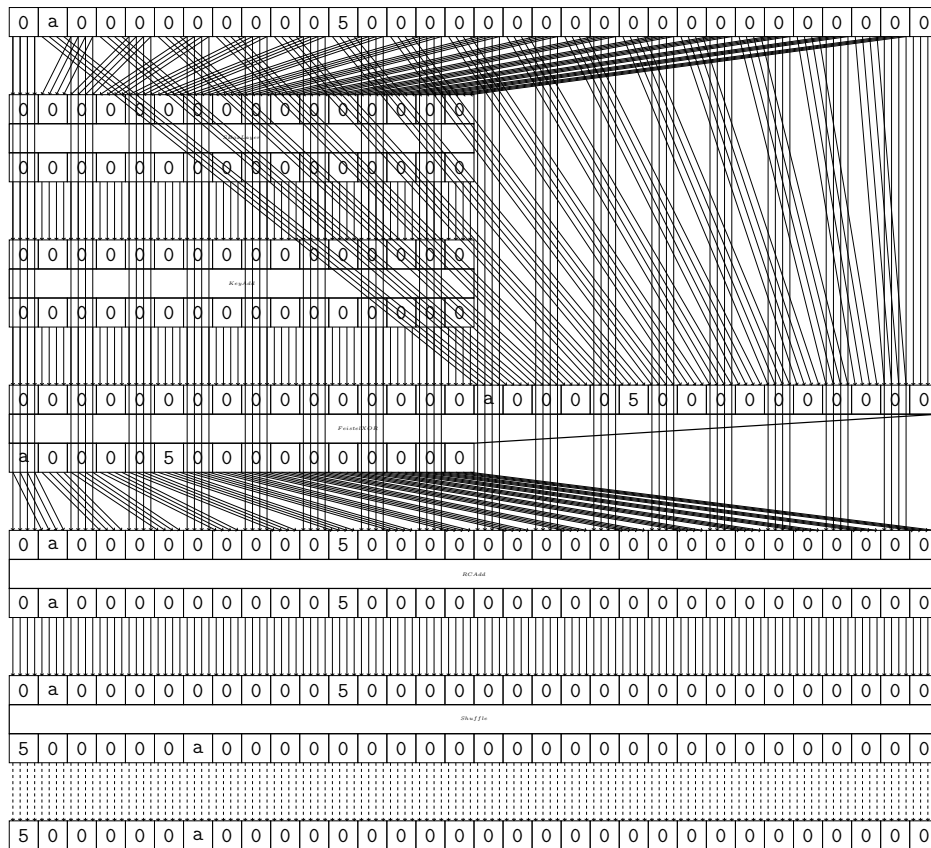


15 SBoxLayer

Maximal differential probability: $2^{-4.0}$

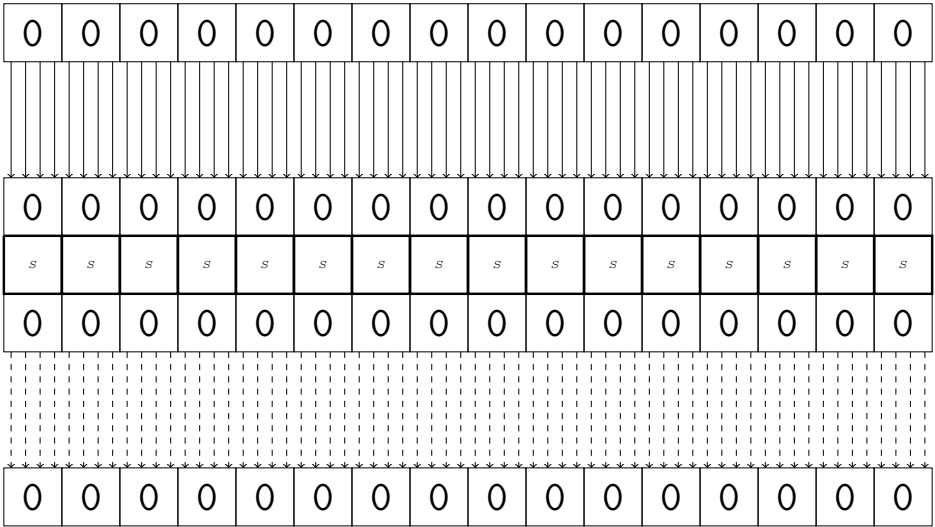


16 warp_round

Maximal differential probability: $2^{-0.0}$ 

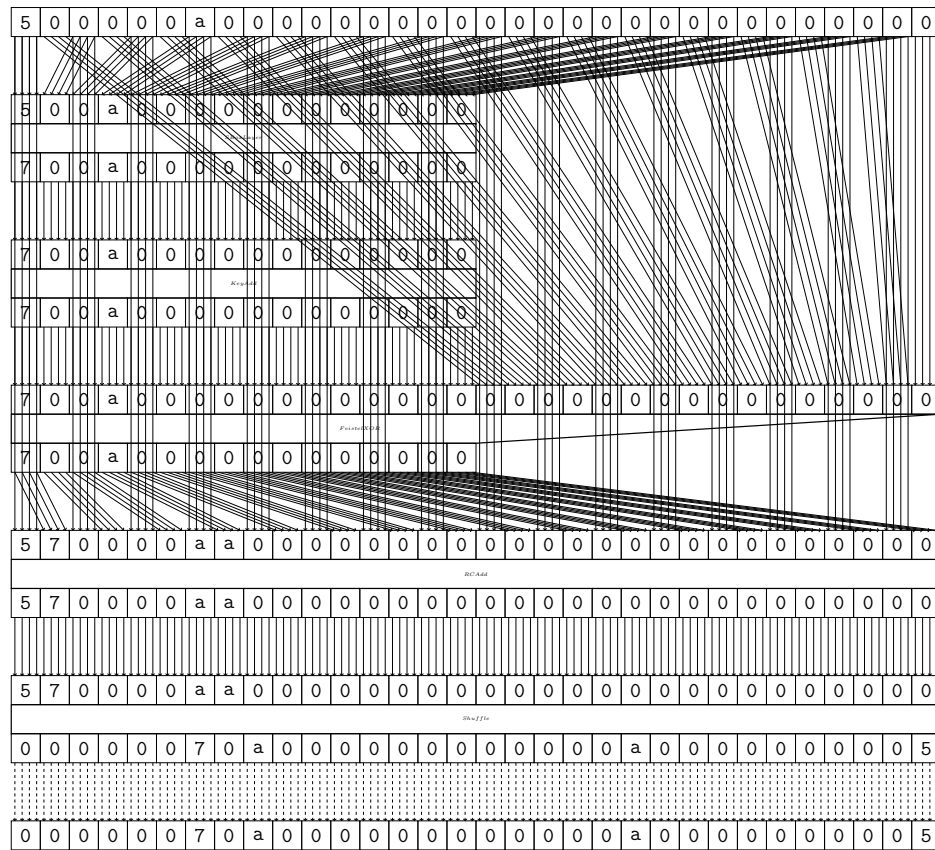
17 SBoxLayer

Maximal differential probability: $2^{-0.0}$



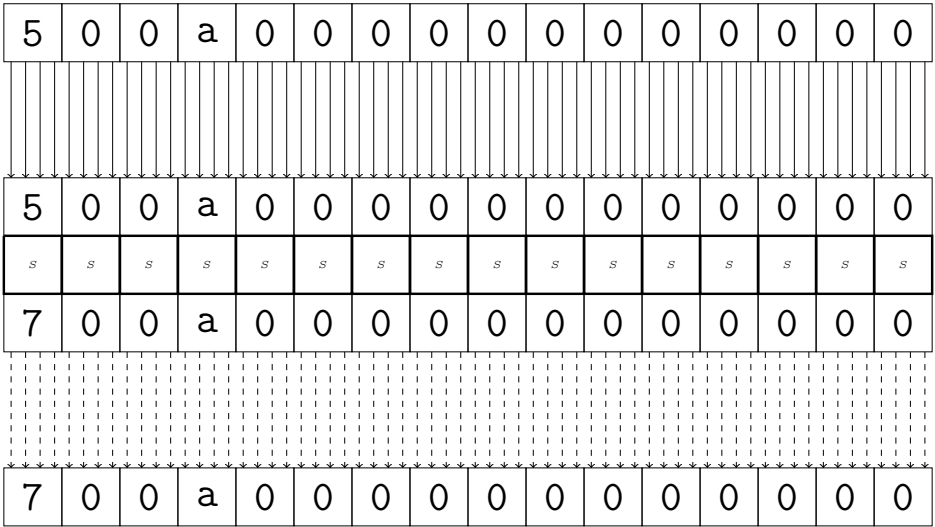
18 warp_round

Maximal differential probability: $2^{-4.0}$

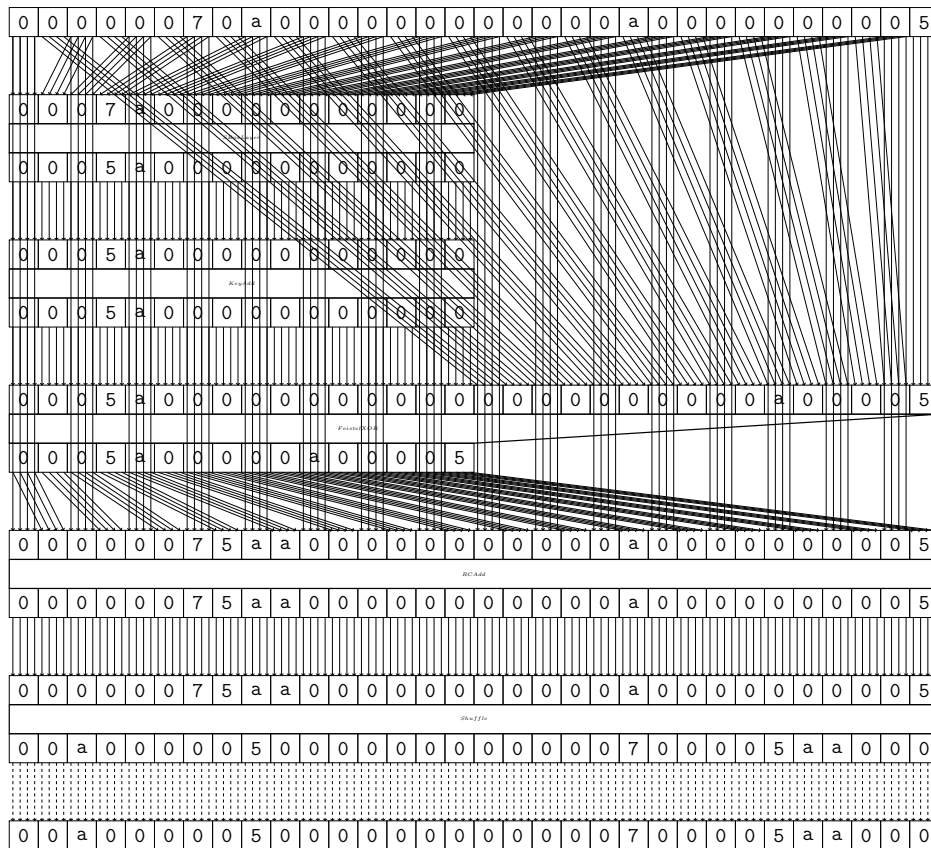


19 SBoxLayer

Maximal differential probability: $2^{-4.0}$

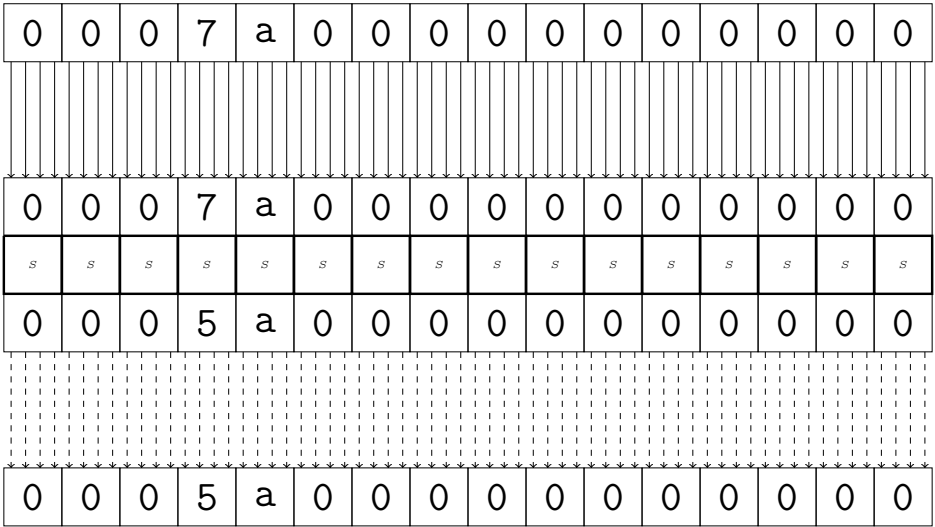


20 warp_round

Maximal differential probability: $2^{-4.0}$ 

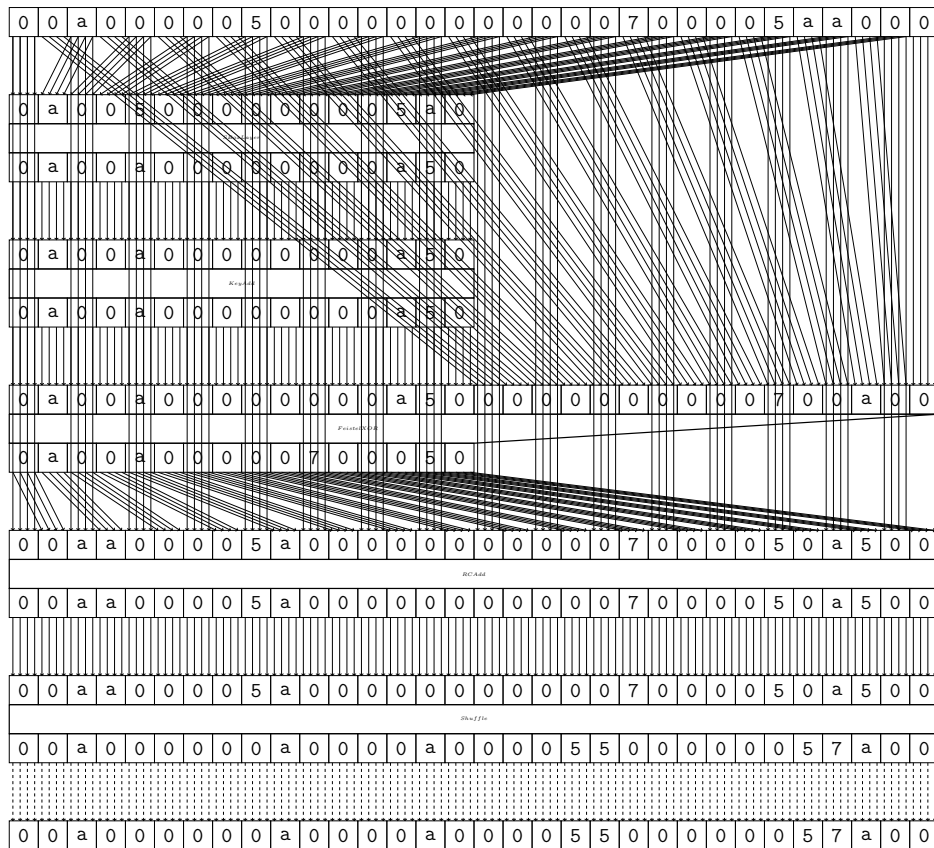
21 SBoxLayer

Maximal differential probability: $2^{-4.0}$



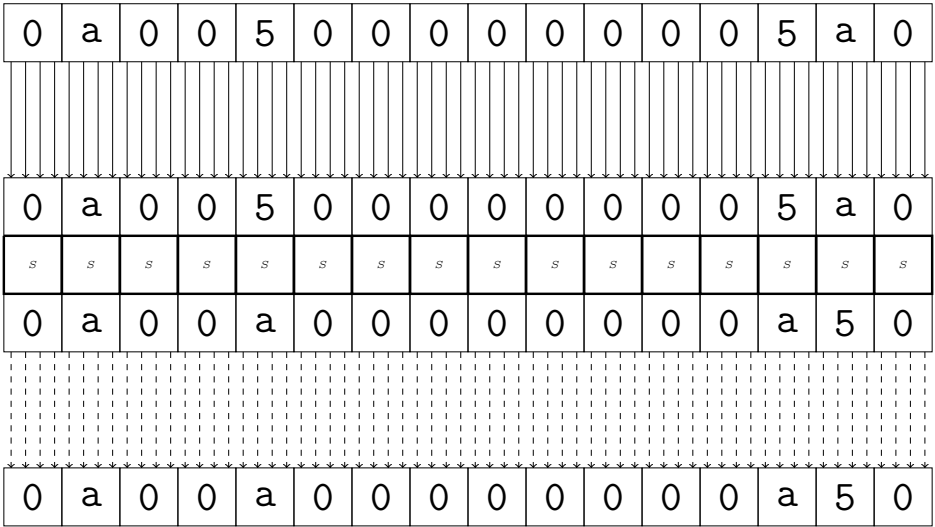
22 warp_round

Maximal differential probability: $2^{-8.0}$



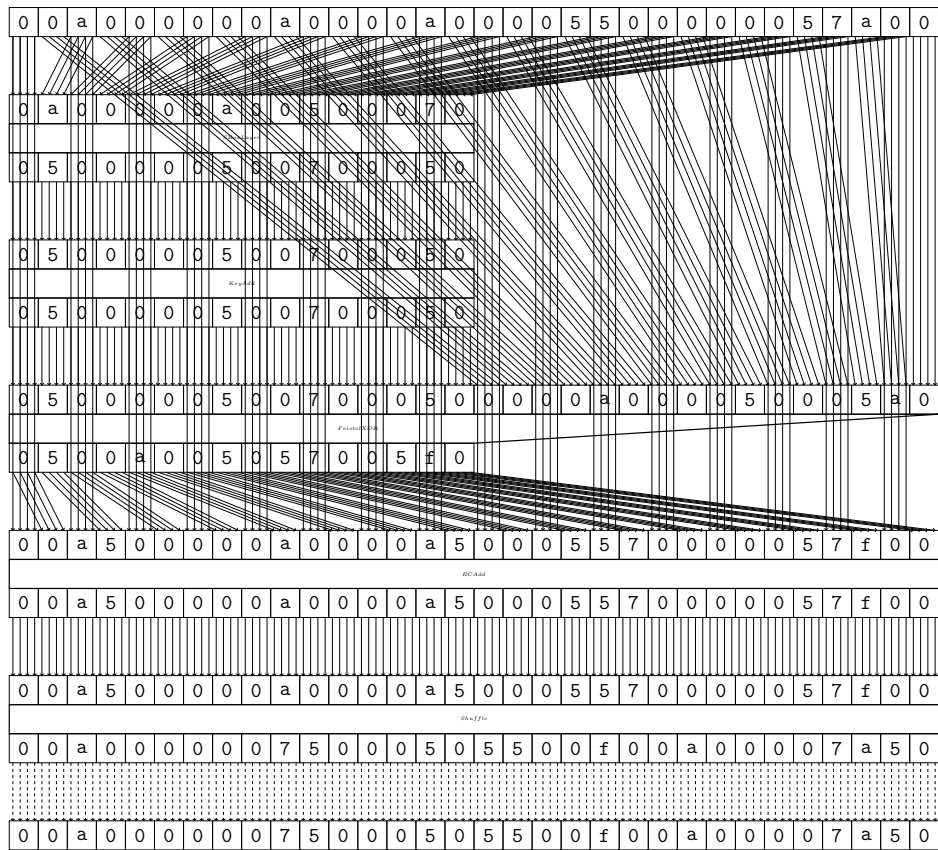
23 SBoxLayer

Maximal differential probability: $2^{-8.0}$



24 warp_round

Maximal differential probability: $2^{-8.0}$



25 SBoxLayer

Maximal differential probability: $2^{-8.0}$

